

# CS 486/686

## Independence and Bayesian Networks II

Yuntian Deng

Lecture 8

RN 13.2 · PM 8.3

Search ›

Uncertainty ›

Decisions ›

Learning

## Learning goals

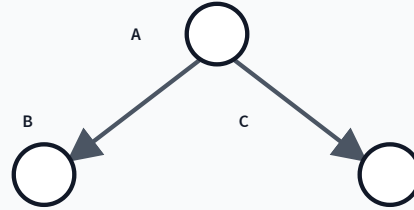
- Apply **d-separation** to check (conditional) independence in any Bayes net
- Given a BN and a variable ordering, **construct a correct Bayes net**
- Distinguish **correlation** from **causation**

# Recap: the three structures from L7

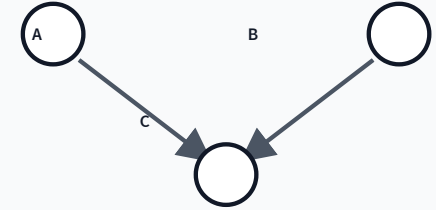
Chain



Common cause



V-structure



Today: generalize the 3 structures to arbitrary paths using d-separation.

## D-separation: the rule

**D-separation.** A set of observed variables  $E$  *d-separates*  $X$  and  $Y$  iff  $E$  **blocks every undirected path** between  $X$  and  $Y$ .

If  $E$  d-separates  $X$  and  $Y$ , then  $X$  and  $Y$  are **conditionally independent** given  $E$ .

A single open path is enough to break independence. We have to check every path and ask: is it blocked?

# The three blocking rules

A path through middle node  $B$  is blocked iff:

## Chain

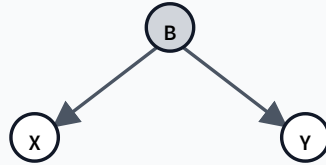
$$X \rightarrow B \rightarrow Y$$



**Blocked** when  $B \in E$   
(middle is observed)

## Common cause

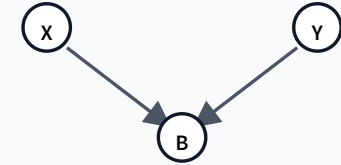
$$X \leftarrow B \rightarrow Y$$



**Blocked** when  $B \in E$   
(middle is observed)

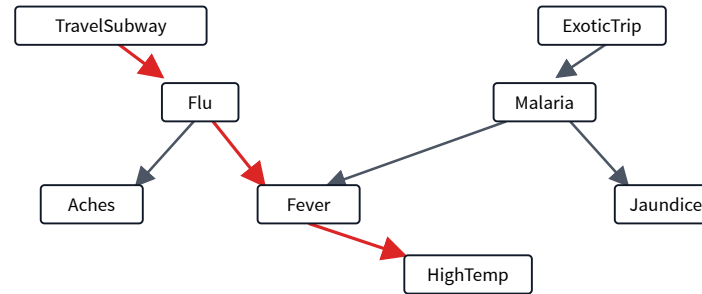
## V-structure

$$X \rightarrow B \leftarrow Y$$



**Reversed:** blocked when  
neither  $B$  nor any of its  
descendants are in  $E$

# Quiz pack 1: chain blocking



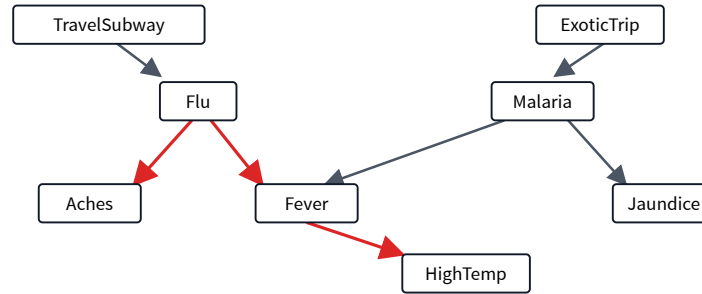
**Q1.** Are **TravelSubway** and **HighTemp** independent?

**No.** The path  $\text{Subway} \rightarrow \text{Flu} \rightarrow \text{Fever} \rightarrow \text{HighTemp}$  has two chain middles (Flu, Fever); both unobserved  $\Rightarrow$  path open.

**Q2.** Are **TravelSubway** and **HighTemp** independent *given Flu*?

**Yes.** Observing Flu blocks the chain at the first middle  $\Rightarrow$  path closed.

# Quiz pack 2: another chain test



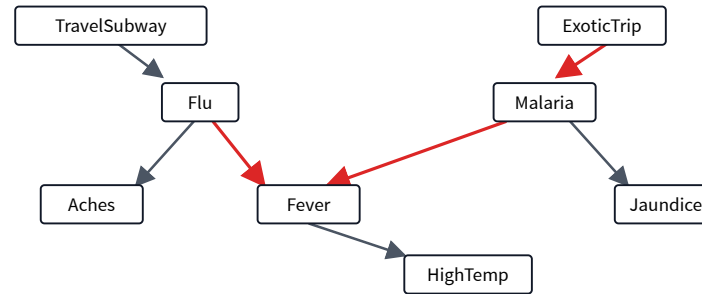
**Q3.** Are **Aches** and **HighTemp** independent?

**No.** Path  $Aches \leftarrow Flu \rightarrow Fever \rightarrow HighTemp$ : common cause (Flu, unobserved) and chain (Fever, unobserved). Both open  $\Rightarrow$  path open.

**Q4.** Are **Aches** and **HighTemp** independent *given Flu*?

**Yes.** Observing Flu (the common cause) blocks the path at the first middle.

# Quiz pack 3: v-structure with a descendant



**Q5.** Are **Flu** and **ExoticTrip** independent?

**Yes.** Path  $\text{Flu} \rightarrow \text{Fever} \leftarrow \text{Malaria} \leftarrow \text{ExoticTrip}$ . Fever is a v-structure middle; neither Fever nor its descendant HighTemp is observed  $\Rightarrow$  v-structure blocks the path.

**Q6.** Independent *given HighTemp*?

**No.** HighTemp is a descendant of the v-structure middle Fever, so the v-structure now *opens* the path (explaining-away).



# Many correct Bayes nets exist

A BN is **correct** if every independence it claims also holds in the true distribution.

## Missing an independence: OK

extra edges — still encodes the joint

## Missing a dependence: NOT OK

too sparse — forces a wrong independence

So prefer the BN with the **fewest edges** (fewest probabilities to store).

# Construction algorithm

1. Order the variables  $X_1, \dots, X_n$ .
2. For each  $X_i$ , choose the **smallest** subset of  $\{X_1, \dots, X_{i-1}\}$  such that, given those parents,  $X_i$  is independent of the rest.
3. Add edges from each parent to  $X_i$ ; write down  $P(X_i \mid \text{Parents}(X_i))$ .

Ordering matters: a bad order can force every later variable to depend on many earlier ones.

# Example 1: chain BN, order $W, A, B$

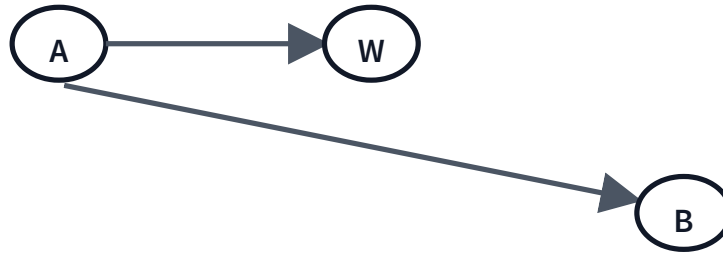
Original BN: 



**Final:**  $W \rightarrow A \rightarrow B$ . Same shape as the original (2 edges).

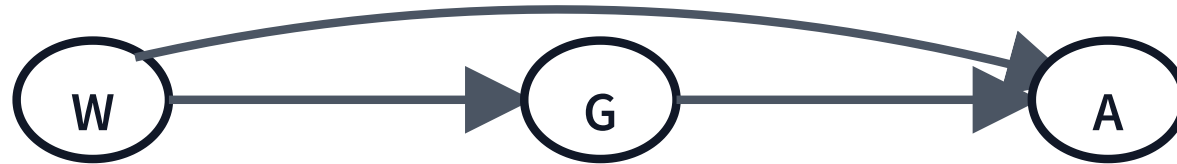
# Example 1 alt: same BN, order $A, W, B$

Original BN: 



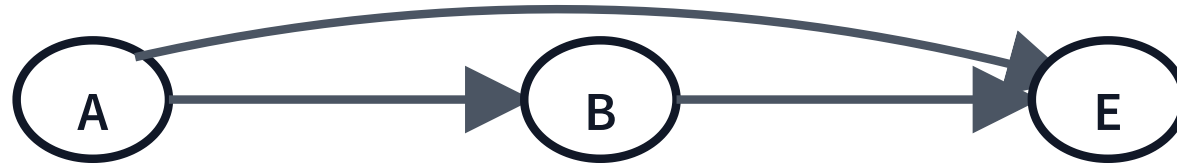
**Final:**  $A \rightarrow W$  and  $A \rightarrow B$ . Different shape, still 2 edges — *different but equally compact.*

## Example 2: common cause, order $W, G, A$



**Final:** 3 edges ( $W \rightarrow G, W \rightarrow A, G \rightarrow A$ ) — *more than the original's 2 edges.*  
Suboptimal!

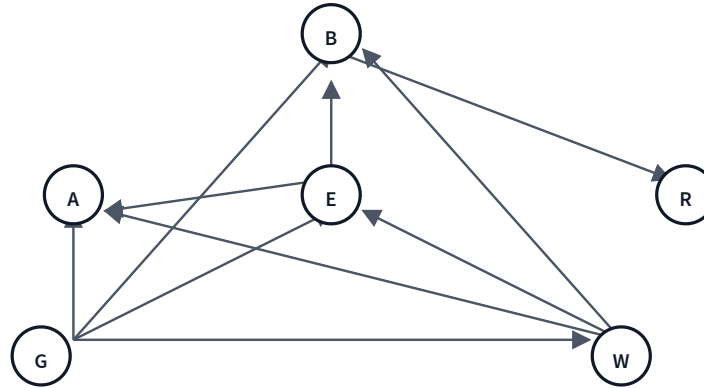
# Example 3: v-structure, order $A, B, E$



**Final:** 3 edges ( $A \rightarrow B, A \rightarrow E, B \rightarrow E$ ) — *more than the original's 2*. Reversed v-structures are expensive.

# Holmes with a bad order: $G, W, E, B, A, R$

Adding effects before causes forces every later node to depend on every earlier one.



$1 + 2 + 4 + 8 + 16 + 2 = \mathbf{33}$  probabilities — vs  $\mathbf{12}$  with the causal order.

# Pick a causal order

Causes precede effects. Add root causes first; effects last.

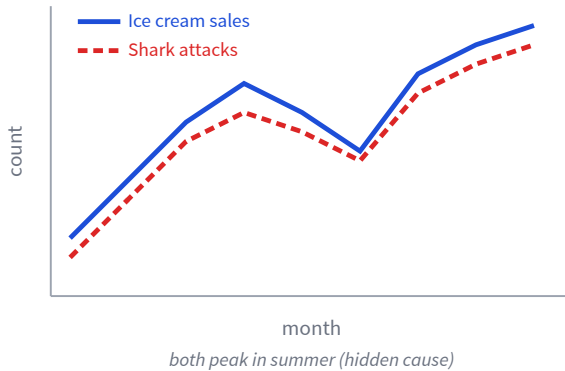
| Example  | Original                           | Order     | Reconstructed                                       | Result                   |
|----------|------------------------------------|-----------|-----------------------------------------------------|--------------------------|
| Ex 1     | $B \rightarrow A \rightarrow W$    | $W, A, B$ | $W \rightarrow A \rightarrow B$                     | 2 edges, same shape      |
| Ex 1 alt | $B \rightarrow A \rightarrow W$    | $A, W, B$ | $A \rightarrow W, A \rightarrow B$                  | 2 edges, different shape |
| Ex 2     | $A \rightarrow W, A \rightarrow G$ | $W, G, A$ | $W \rightarrow G, W \rightarrow A, G \rightarrow A$ | 3 edges — worse          |
| Ex 3     | $E \rightarrow A, B \rightarrow A$ | $A, B, E$ | $A \rightarrow B, A \rightarrow E, B \rightarrow E$ | 3 edges — worse          |

Finding the most compact BN is NP-hard in general — but a causal ordering is a good heuristic.



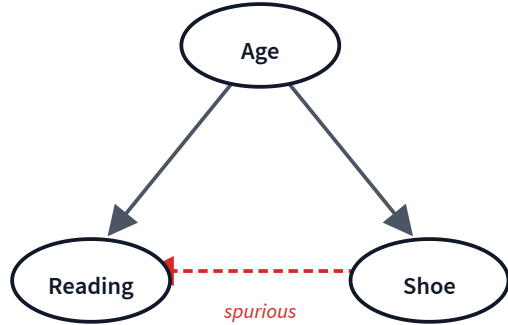
# Correlation $\neq$ causation

Two variables can be highly correlated without one causing the other.



- Ice cream sales correlate with shark attacks. Does ice cream attract sharks?
- No — both rise in **summer**. Temperature is a hidden common cause.
- An edge in a Bayes net is associational, *not necessarily causal*.

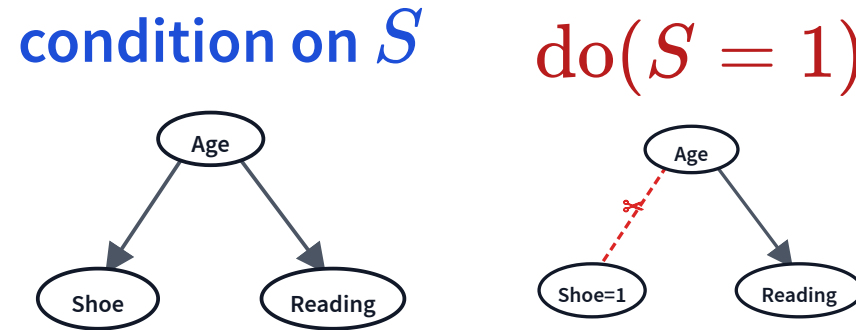
# Confounding variables



- Bigger shoes correlate with better reading scores.
- **Hidden cause:** Age. Older children have bigger feet *and* read better.
- The Shoe → Reading "effect" is **spurious**; controlling for Age would erase it.

# Causal intervention: the do operator

- Condition  $P(R \mid S=1)$ : what we see — large feet correlate with reading.
- Intervene  $P(R \mid \text{do}(S=1))$ : what happens if we *force* big shoes on a random child.



Intervention *severs* the incoming edges of  $S$  — the confounder can no longer reach  $S$ .

# Adjusting for confounders

The **average treatment effect** averages the effect over the confounder  $A$ :

$$ATE = \sum_A P(R \mid S=1, A) P(A) - \sum_A P(R \mid S=0, A) P(A)$$

$$ATE \approx 0$$

Shoe size doesn't *cause* reading skill — randomised experiments confirm it.

## Learning goals (recap)

- ✓ Apply d-separation to check (conditional) independence
- ✓ Construct a correct Bayes net from a variable ordering
- ✓ Distinguish correlation from causation

## Next: decision theory

We can now reason under uncertainty. Next, we combine these probabilities with **utilities** to *act* — choose the action that maximizes expected reward.